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MAY, 1955

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by Ray C. Noll



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*** START OF THE PROJECT GUTENBERG EBOOK FLIGHT
PERILOUS! ***

Flight Perilous!

By Ray C. Noll

**As Captain of the ship, Hiller knew full
responsibility was his, if he ordered Marship III
through the uncharted asteroid belt—to death!...**

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As Fred Hiller slid back the door to his quarters in answer to their knock, he found them lined up tensely against the bulkheads of the companionway.

It was the best assembly area the jammed ship could offer. Here the commander with a short turn of his head could meet any pair of eyes in the nine-man crew. They had met here before, in a more friendly atmosphere, soon after acceleration stopped and once for planning. He considered it more effective for personal communication than the ship speaker system.

But this assembly was different: it was their idea. They wanted a decision. They stood without moving, waiting for him to speak. Their discussions by this time probably had narrowed the alternatives to two.

As commander, of course, he was paid to make decisions on Marship III. And he began to realize by their faces which alternative the consensus expected. Their expressions indicated that in a degree every damn one of them was scared, scared enough to unitize their thinking.

Phil Bleck was the one fishing for an impressive opening. He moved forward to face the ship commander with hands pressed on his hips defiantly. This was *the* Phil Bleck, young man genius of United Nuclear, pressured aboard Marship III as nuclear engineer through a couple of Senators and the Secretary of Defense. Oh, he was good, as long as he wasn't under fire. So good posterity required him and he was obligated to save his skin. Hiller had expected Bleck would be the spokesman.

"We want to know if you decided yet, Hiller," Bleck nearly mocked.

"*I'd* have called this assembly if I had," Fred Hiller replied, emphasizing a commanderish tone of voice.

"Then you haven't." Bleck turned to the others significantly and brought back with him a harsher gaze, which he leveled at the commander. "Most of us here think there's only one sane way out. A couple will go along with any decision. But most of us, including me, want to turn back. Isn't that right?" He turned again to the men for support. Some nodded.

"We figured the chances if we keep on course," Bleck went on, breathing a little heavier. "They're three to one against making it. I don't like those odds, Hiller,"—his upper lip was curling a little—"and we didn't agree to odds like

that when we volunteered. With what we know now, we can plan another trip and avoid this mess next time. That way, you'd only waste time and money; going ahead, we waste that plus the priceless knowledge of these scientists, the best the States has to offer."

While Bleck was blowing off, Hiller had studied each man in turn. They hardly represented a crew, though the men had specific jobs to perform during takeoff, transit, and setdown. They represented specialists who would bring back for the first time authoritative reports on Mars—the first two ships had not returned....

Marship III, several times the size of the first ones, but not one-hundredth as much publicized, had been under construction since the first Marship attempts.

The crew technicians Hiller possessed on the trip were three. And as he found the eyes of each, he realized they were not with Bleck.

Art Eastburn, an all-around engineer, whose capacity continued to amaze Hiller, and who had helped build the Marships.

Dave Hollender, astronavigator, bucking for a space ride ever since the moon-missile days; a cool thinker, who had the solar system duplicated and in accurate motion inside his skull.

Wendell Merrick, electronics engineer, who supervised the wiring of Marship III and was sensitive to the click of every relay in the almost fully automatic craft.

These were with him, which fundamentally meant they were willing to continue on course if he so decided. The others had succumbed to fear, and they recognized no authority nor purpose: their choice was a reactionary Earthward course.

"Dammitohell, Hiller, we want an answer!" The commander's silence had edged Bleck better than words. "The issue can't be plainer. Let's get this indecision over with and give the orders to circle back! Or do you want us to end up as dead as the first two Mars attempts!"

"Again, Bleck, I haven't decided," said Commander Hiller coolly. "I'm going to take more photographs with the Newtonian. What comes out of that will affect any decision I have to make. But since we're so concerned with decisions, Bleck, have you decided what you'll do if I should order us through?"

The commander's unexpected and pointed directness left Bleck blinking long enough for Eastburn to cut in before any heated rejoinder by the young nuclear engineer. Eastburn, because of his prominence and experience, held the respect of most of the men.

"When I volunteered for this jaunt, I also agreed to follow the commander's orders," Eastburn said firmly. "He may be wrong, but I could just as well be wrong in thinking he is. We're after unity of action, so at least something gets done in some direction."

Hiller smiled inwardly at that choice gem of rationality because the crew's emotional perception made of it no more than a granule of gravel. They would have to be appealed to emotionally; under the pressure, they understood nothing else. The stir of resentment evoked by Eastburn's words was dying down.

Bleck had started to say something, but Hiller's voice drowned him easily with its overpowering bass.

"Then, let's put it this way. Suppose I decide to hold course and you—ah, let's say—'persuade' me to circle back. When we all testify at the hearing, I hope you don't expect me to protect you. I'll tell them exactly what was behind the mutiny, your yellow vertebrae, and what would that do to your reputations?" Hiller had to shout the last words, because Bleck was screaming interruptions.

"It's your word against ours!" yelled Bleck into Hiller's sudden silence. "It's your word against ours that you didn't crack and blame it on us!"

The commander lifted his eyebrows. What perfect projection!

"I guess somebody in a spot like this could crack, couldn't he?" Hiller purposely addressed the remark to Bleck's followers. Most of them were staring uncertainly at Bleck's perspiration-soaked shirt, his white face, the hunching shoulders, and moving wordless lips.

"For the time being, let's leave it this way," said the commander authoritatively. "Unless conditions improve, we're turning back. If the odds seem later about even, we're going through. In the meantime, we'll make these preparations just in case we can chance the clusters."

Possibly the instructions he gave sounded casual and spur-of-the-moment; actually, they were the careful product of his close figuring and planning, made during the last eight hours. It was more a recitation, yet he had to make it seem ad libbed. No one yet knew he had resolved on what data he had at present to hold the ship's Marsward course.

Even as he energized the lock mechanism on the door of his quarters, Fred Hiller began to tremble, a violent physical reaction of taut and unrested nerves. It had been capped by the crisis of the crew's resistance, a matter hardly settled, mainly delayed.

He fell into his bunk and let the shakes take over. Right then they felt ghastly, but he realized he'd feel better when they stopped. As they subsided, he tried to keep the problem out of his mind. He was too tired for that; the pictures returned again and again in front of him mostly beyond his control.

He stopped fighting them, and let the pictures progress. He justified the surrender with the thought he might learn something, might conceive a better protective device against the myriad missiles of the Belt.

The same picture always started it—Lord, was it only a few hours ago?—when Dave, the ship's astronavigator, called him to the observation bubble....

Dave spent his time at the compact reflector, peering into his frequently changed eyepieces and setting up one photograph after another. The instrument was his own design, with a revolving optical flat tempered for space temperature that could be suspended out from the ship and rotated, effecting nearly a 270-degree field for the telescope.

"Take a look," Dave said. At the time, he thought there had been a slight edge to the astrogator's voice.

"Don't tell me you brought me up here again to admire colors in another variable," he had grumbled.

"You won't admire this a bit," Dave replied.

"Where're we looking?" he asked, slipping into the seat behind the eyepiece assembly.

"Space," Dave murmured. He was sighting in the finder and made azimuth adjustments.

When the field slid to rest Hiller viewed once more the gripping vastness of black wantonly perforated with intensely glaring stars. It was impossible to study the closer ones; their brightness and energy coursed pain along his optic nerve. Rather, he let his gaze wander over the distant sprinkling of light that marked milestones toward infinity.

"Notice that hazy part in the upper field," Dave was saying.

He found it, a faint stellar gauze wisping before the stars. It appeared to be moving. But that kind of rapid movement was out of the question; it would have to be too close.

"Now, I'm tripling the power," the astrogator informed him.

With the new eyepiece in place Hiller noted that the haze had condensed into fine dust, each particle of which contrasted dimly against space compared to the stars over which it was super-imposed. And it *did* move! Part of it already was creeping into the invisible curve of the eyepiece rim.

He pulled back from the telescope to look at Dave's grim features. The quickening in his stomach meant anxiety, he recognized it easily. Anxiety over what? How could he have known then what it meant? Subawarely, he must have.

"What the hell is it, Dave?"

Hollender handed him photographs out of a transparent file cabinet. "Here're some blowups under high power. Visible proof from these, but nothing highly accurate from the spectography."

"But this is asteroid stuff," Hiller nodded at the photographs. "They look like pinhead star clusters."

"That's what they are, clusters," Dave replied seriously. "Fragments of planetoids, evidently, revolving around common centers of gravity."

"What're they doing around here? I mean, are they strays from the Asteroid Belt or something?"

Dave swept the back of his hand over a row of astronomical texts. "If you can find any mention of 'em in there, I wish you'd let me know. And they're a long way from the Asteroids."

"Well, you're the damn astronomer in this blowout," frowned Hiller impatiently. "What's a good guess on 'em?"

"I don't know how good it is, but my guess is we're running into an inner Asteroid Belt. I'll bet the first two flights ended here...."

"A *what*?"

"Inner Asteroid Belt," Dave stated. "A puny one, compared to the one outside Mars, but nonetheless a Belt. Uncharted, and deadly."

Tension spread along Hiller's back. "We've had no advance data on crap like that, not one bit."

"We have now," Dave shrugged.

"Why didn't one of the first trips miss this?"

"This Inner Belt orbits, too. Clustered minor particles with low reflectivity may be a phenomenon found only in scattered sections of the Belt. The first Marships happened to hit them, just like us. Certainly no light instrument on Earth I know of could pick them up. They'd move out too fast to register on a plate. So, they're our babies."

The commander remarked soberly, "You evidently already know what this means."

"It's a lousy break.... The first ships must have tried to get through...."

Hiller brought his palms together to bolster the searching of his mind. He was surprised to find them moist. "What about the size of the particles in these clusters?"

"From what I've calculated, they're fruitstand variety, for the most part."

"Watermelons?" grunted Hiller, pessimistically.

"A few, maybe. But they're not cranberries, either."

"Density?"

"Roughly eight or nine. I can get that figure closer later on."

Hiller became irritated at himself for letting what started to be short silence grow longer. The astronomer may have followed his thoughts; he handed him a long photograph.

"Here's one I made at 150 diameters of the general area of the Inner Belt we're due to pass through on our present course."

Hiller winced at the sight: the fuzzy glow thinned in the foreground and thickening, paraded through the middle distance, still stretching on until it faded from the lens' capability.

"We'll have to revise some of our theories about the formation of the Outer Belt," Dave was saying. "It's apparently much deeper and wider than anyone's guessed. Looks to me like a dead star went through our system, breaking up a planet and maybe peeling a little off itself. That would account perhaps for the retrograde orbit—"

"Dave, I don't give a good goddam about any dead star!" Hiller exploded his tension. "How far apart are these space fruit?"

"A mile here, a couple of miles there. I really haven't figured that aspect yet."

"Well, figure it." The commander jumped down from the observation seat. "Get George, the psychologist, he types fast. Compile what data you have, have him type it, send it down to me. I'll be in my quarters. And hurry, man, or they'll be more than stars dead around here."

He slammed open the entrance panel to the observatory. By that time he had cooled enough to pause and throw Dave a half-smile and limp salute.

"Thanks for the wide-awake work. Now, get busy."

His watch showed he had been drowsing for more than an hour. The pictures had exhausted themselves, and his head felt clearer. He had needed that rest badly.

Sitting up, he reached into the bunk cupboard and poured a drink. Now that Bleck was temporarily emotionally neutralized and the brains uncertain, it was time to follow up with a little rationality to substantiate his position. Anyway, he wanted verification and cross-checking of his plans. He *could* be way off base.

Over the ship's speaker system he summoned Merrick, Eastburn, and Hollender to his quarters. They arrived promptly, almost too promptly, as if they had been waiting. It was probably obvious to them, as it was to him, the problem called for more than one man's calculation.

Nothing was said while he splashed out drinks. The men spread over the floor where they could find room and left him the bunk. They were evidently going to let him say something first, so he didn't disappoint them.

"I don't think I'm surprising any of you when I say we're pushing through the clusters, regardless of Bleck's nerves," he began. "What's probably on your mind is my motive. You may understandably feel Bleck, no matter how badly he expressed his point of view, may have something. Sure, maybe my pride is driving me ahead. Maybe I'm being as emotional in wanting to buck the clusters as Bleck is in wanting to run.

"I'll let you judge that for yourselves after you hear what's back of those orders for preparation I gave. First I want to hear from Hollender. What's the latest and most accurate you can give me now on density of the particles, particle proximity, and our relative velocities?"

The astronavigator unfolded a paper taken from his shirt pocket. "Well, I have three results on density because of observation problems. I'll give you the average. Mean density comes out to 7.8, lower than I first figured. Roughly on proximity, 1800 yards, and that's more bunched than I estimated. They're clustered, and that's about it," he shrugged.

"Now, on relative velocities," he continued, "I could get it pretty close, knowing ours is a constant power-off glide. We exceed the clusters' orbital velocity by three m.p.s. But our angle of intersection with the Belt will

reduce any actual impact to about two m.p.s. In other words, particles would be overtaking us at about that speed."

Hiller nodded. "That's about the way I worked it out. One more thing, Dave: the depth of the cluster band."

"The part we have to worry about's only a little over a hundred-thousand miles in depth. The rest is scattered asteroid strays and shouldn't bother us. We'll be three hours maybe in transit through the stuff."

The men in the cramped commander's quarters stirred slightly, wincing at the transit time. The other figures could not be readily personalized; but each of them could visualize himself sweating out three hours of stellar bombardment, the effects of which would not be known accurately until the Belt was entered. And each could visualize ultimately Marship III as a whirling, shredded mass, spouting synthetic atmosphere, and glowing redly from rampant and uncontrolled fusion.

"On the fuel?" the commander asked of Eastburn. "Anything new on that?"

"Deceleration definitely out," the engineer replied firmly. "We couldn't afford the drain needed later to catch Mars on her way around. From what I gather of the problem, acceleration wouldn't do anyway, but that's even more impossible. It would increase setdown consumption."

"Hollender and I've calculated the fuel drain required to circumnavigate the clusters. It came close, close enough to make you want to cry. But not close enough. The wall of the clusters happens to be too spread out and in near-perfect line with our point of rendezvous with Mars. If we'd spotted them sooner, we could have hurdled 'em with a few spurts of the guide jets. By the time we got it figured, we'd already passed the critical point by 23 minutes. That's how close it was.

"The fuel was figured for this trip with very little margin, and we used some margin already because of that lovely instrument error on takeoff. I'd be a lot happier if we had a fusion system with fewer limitations, like the ones they're working on now."

"We agreed to this firing system and realized its risks—all because we'd rather not wait for the ones in development," Hiller reminded. "We're comfortably powered, anyway, provided we follow our original firing schedule. So, that means we enter the Inner Belt at our present velocity without changing course."

Merrick spoke up, ruffling the red hairs that partially covered his shiny scalp. "Back track here a minute, you boys went over that rapid. I think I get everything but the velocity business. We connect with the Belt at two m.p.s.? Sounds like optimistic but bad arithmetic to me." Screwing up his mouth, he squinted at Hollender.

Hiller found himself laughing, and it felt good. "Pardon our dynamics-centered minds," he said. He unsnapped his ballpoint from his pocket and placed it over the air blower grill.

"Say the horizontal braces on this grill running parallel are the clusters' paths at 12 m.p.s. My ballpoint's the ship at 15, traveling in the same direction as the clusters. In that case, we would collide with the particles, overtaking them at three m.p.s. right?"

Merrick nodded. "I see that, but—"

"Okay," Hiller went on. "Now suppose we crossed the Belt at right angles to the paths." He moved the ballpoint straight up the grill.

"They'd sock us at 12 m.p.s.," Merrick deduced. "So, what you're getting at is the angle—"

"The angle makes the difference," finished Hiller. "If we entered the belt at about this angle"—he inclined the ballpoint up slightly from the horizontal—"we'd sail through with the same velocity as the particles. If we hit any, it would be a nudge from our transit motion through the Belt or from their velocity or revolution, which is probably very low."

"I get it," Merrick slapped his forehead. "Our present course cuts the Belt at such an angle that we get bumped at two m.p.s. instead of 12." The others nodded. He reflected a moment, adding, "So, I get a bullet through the head at 1200 feet per second or 200 feet per second: I still get it in the end."

"Not in this case," Hiller smiled. "There's been quite a little work done on effects of meteor impact by the Air Force. I've got a summary of it in the control room. Art, here, could probably tell you more about it than I could."

Eastburn hugged his knees. "Not much, I don't think," he arched an eyebrow. "Fred's being modest, the guy who designed the meteor-scanning device used on all Marships. I'll take the ball, though, on this one."

"Del, we've got a brute of a hull on this ship, twice as resistant as the ones on I and II. Second, it's smooth and curved. Third, it's going awfully fast. The studies the Air Force has been able to make so far show that small-sized meteors either glance off a ship and disintegrate swiftly from the excessive rotation set up from the collision, or they explode on contact from built-up kinetic energy."

"There seems to be three types of contact explosion. Where the angle of impact is not quite perpendicular, the particle creases the hull and explodes along its trajectory. This is the usual situation in the heat-generation collision and rarely harms the ship."

"Perpendicular impact, however, does the damage. At low velocities and densities perpendicular impact craters the hull and most of the blast effect is dispersed laterally and to the rear. At higher velocities the particle vaporizes but the explosive force craters the hull and shapes inward, a lot like the effect of an air gun pellet on plate glass. Although the hull penetration may be mere pea size, blast and compression inside can be terrific, besides the sharp shock throughout the whole ship."

Hiller grinned. "Thanks for bringing out the situation so well, Art. Hollender's the mathematician here, and I don't go in for formulating odds. But I'll give odds right now on our getting through with one perpendicular strike. Any takers?"

"That's a hell of a bet," Merrick griped. "If you lose, who's around to collect?"

"I'm talking odds," the commander said. "Anyway, you over-estimate the effect of a perpendicular strike. In a closed compartment it could be rough. By leaving every compartment hatch open, the compression would dissipate throughout the ship with less damage."

Art Eastburn frowned. "How about the air supply, Fred? With no compartmentations, one big enough hole and most of the ship's air supply

could escape before we could patch up."

"Good point," Hiller replied, "but if the hole were as large as you may be imagining, the blast pressure would probably blow out ports and open seams, leaving us in hopeless shape. The smaller holes, on the other hand, could be patched, the kind we expect. I have reason to believe that won't be a problem. A hunch, maybe."

"I guess we can let you get by with one hunch," Eastburn smiled wryly. "But I can see what you're getting at on the odds you mentioned. Considering Hollender's estimates on the spacing of this fruit-sized stuff, I might not take your bet."

"Another factor," the commander noted, finishing his drink. "You don't go through a barbed-wire fence standing up."

"Granted," agreed Merrick. "Are you giving again with that ballpoint?"

"Last time," promised Hiller. He held the pen over the grill, pointing it at the approximate angle the ship was to take through the Inner Belt. "That's the way we're heading now. We've set the gyros to keep our nose in front, for the time being, to satisfy tradition and maintain a consistent sighting base."

"Our main problem is avoiding perpendicular strikes and encouraging oblique ones. The position of the ship in relation to the particle direction becomes important, then." He moved his ballpoint at nearly a 45-degree angle to the grill lines. "We won't head in the course indicated by the nose, but we'll gyro the ship to this position. That way we obtain the maximum deflection."

The men were silent momentarily. Merrick suddenly sat up straight.

"It seems to me pointing the nose right at the asteroid flow would be better."

"You forget our transit velocity, Del," the commander observed. "We'd be chancing running into as many particles perpendicularly with the ship lengthwise at two m.p.s. as we would miss by pointing our nose at those catching up with us at about the same speed."

Merrick threw up his hands. "Okay, okay," he surrendered. "All I hope is you math boys have it figured right."

"We're running it through the calculator to round off the rough edges," Hollender assured him.

The silence grew until the commander stood up and asked. "So, on the basis of what we've covered, am I too much of a gambler in going ahead?"

The others had risen and Eastburn was the first to offer his hand, the others following. They spoke at the same time their assurance and backing. But Hiller's thoughts were already dwelling on the most bothersome variable of all—Phil Bleck.

When he discovered from Hollender before he left that Bleck had no idea when the ship would enter the Inner Belt, the variable began to assume minor proportions.

"Test drill 30 minutes! Test drill 30 minutes!"

The commander adjusted the mike closer to his chest and turned up the volume on the portable transmission unit for the ship speaker system. Under the coming circumstances he would need as much freedom as possible.

The panel before him gradually lighted up as the stations checked in. They were in no hurry since he had informed them in the last meeting that the Inner Belt was still six hours away. That had provided Bleck with enough time to map what counteraction he had in mind to oppose a decision for continuing Marsward.

The commander noticed with satisfaction the colored lights wink on over the board, each with its own vital significance. The row to the left on the panel, half alight, indicated locked-open compartment doors. Near the bottom a circular array showed Eastburn was prepared to activate the gyros from the mechanical control center of the ship. The green bulb newly burning indicated Merrick had completed his check of the electronics at the control center in the next compartment to the commander's and was standing by.

The blue glow at the top of the board was Hollender at his observation post. The fire control posts—two, stationed near the ship's center—blinked in almost together. Wayne Somerset, chemical engineer, headed the patch crew made up of the zoologist and archeologist, the team which was the last to signal readiness.

It lacked 12 minutes until drill time.

Hiller switched on the monitors for the nuclear chambers which he lighted up by activating remote spotlights. He had some trouble adjusting the scanning in one of the monitors for the fuel compartments, but it came in clear by 10 minutes until drill.

"Test drill 10 minutes!" he announced. "I want an oral report on these items from your stations: suits, rations, extra oxygen portables, first aid and anti-ray kits."

The reports came in affirmative, and Hiller relaxed slightly. The phrase "shipshape" kept coming into his mind but he rejected it as histrionic. But maybe that was the word for the whole situation, with his being guilty of plenty of hamming. Come to think of it, it was more like TV fantascience than anything else.

"Bleck," he broadcast, "leave George at the fire station and report for special orders."

He suspected Bleck was sulking through the preparations and would do George little good. The best place for Bleck was with him, suspecting what he did about the man's reactions.

"Test drill five minutes," he was announcing as a sullen Bleck arrived at master control.

"Art, better adjust the pumps to lower air pressure. Somerset, plug in the patch kits for molten. Fire crews, uncap and pressurize your mist tanks."

Hiller swung in his chair to face Bleck. "Sit down," he said. He caught the puzzlement on the man's face over the realistic degree of the last orders he gave.

"Adjust the magnetizing on your boots to high, unless you have to travel," he continued. "Unbind emergency deceleration straps and stand by."

Bleck's color faded with the commander's last words. "Why the hell all the realism, Hiller? Your rank puffing you up?"

Keeping his eyes on Bleck, the commander went on, "One minute to test drill. Only this isn't a test drill. Repeat, this is *not* a test drill. It's the real thing. We are now into the Belt. Repeat, this is the real thing."

Bleck clawed over the bulkheads of master control's cubicle searching tactilely for the deceleration straps, his eyes riveted blankly on Hiller.

"I take complete responsibility for this deception," Hiller spoke to the crew, "and I can justify it. Yes, Hollender, Eastburn, and Merrick were in on it. They also agree with me that our chances of getting through are good as long as everyone does his job. You should be glad I saved you worrying.

"We're inside the Belt now and the way to get out alive is to stay alert and follow the drill plan. I'll keep you informed from master control how we're doing without pulling punches. Let's have nothing on the intercom unless it's strictly business."

Bleck had found the straps, but he had not fastened any. Instead he crouched, burrowing his head into one of the pads. He was curling up in a knot and sobbing.

"I figured you'd break," Hiller mumbled more to himself than to the quasi-comatic nuclear engineer. Breaking, this was the best place for him. He wouldn't exactly boost the others' morale were he around them. Nor with Hiller's dirty pool, could Bleck get the chance now to lower morale enough to push over a mutiny.

"Art, let's gyro her to the transit angle," he broadcast. "I'll cross-check on my indicator up here."

A faint vibration seeped through his feet as the electric motors revved. Watching the unmoving star-scape through the front ports, he waited for the slow shifting of the field. The effect was as if the heavens had begun an expansive revolution about the ship, the stars drifting lazily from their familiar positions in the ports.

The commander watched the positional needle creep away from the arbitrary course zero. It swept beyond 10 and slowed at 15, halting a little beyond 16.

"I show 16.2," Hiller communicated.

"Check," Art answered on the intercom.

There it was, physically as much as any commander could do under the circumstances. The rest was largely luck—and, of course, how fast he acted to offset any bad luck.

Hiller took the time to explain to the crew the tactics planned in traversing the Belt.

"You guys are gamblers or you wouldn't have volunteered for this commute," he concluded. "The only difference with the hand you're holding now is that somebody else had to figure the odds for you. They're not bad odds either. If you grouse and jump for the straps every time a plum taps the hull, they're 50-50. Keep your heads and follow my instructions and the odds go in our favor.

"We're going to be hit, we're going to be hit again, and maybe a couple of dozen times after that. If a big one slams straight into us, somebody might get a bloody nose. But we can get through even if the ship turns out to look like a thick piece of Swiss cheese.

"Right now we're sailing in between thinned-out stuff, Hollender tells me. The first hour will be a tea party compared to the second.

"The air pump room sits smack in ship center. Anyone who'd like to zip his suit and shut himself in with the pumps has my permission. Speak up now; I can't force co-operation in something like this."

The intercom stayed silent.

"Thanks," the commander said. "One more thing. Fish a couple of hunks of cotton out of your first aid. After you hear the first hit, you'll know where to put 'em."

Hiller watched the changed stellar configurations through the ports. The stars shone in a friendly brightness compared to the darkness around them. That darkness held invisible missiles which possessed only velocity and direction, harmless vectors. Only when they met the hull would there be a molecular and not only calculative indication of their presence.

The ship rode silently, weighted with the heaviness of a grim expectancy.

Hiller curiously switched on his meteor-scanner, making sure to keep the circuits connecting to the guiding jets and gyros cold. Even if they could

afford the fuel, the gadget would tear itself apart with the plethora of loose particles to monitor.

The greenly glowing two radar scopes' limited field was clear for the first few moments. Then three fine lines sped down the center, and before they faded two others plummeted beside their fading tracks.

Watching the scope fascinated him. The lines traced, glowed, and faded, always cutting the same angle, so far staying fairly clear of the center. He caught himself tensing when one began at top center and coursed swiftly toward the ship. A trail actually disappeared under the center marker but came out the other side too swiftly for him to wince under the anticipated shock.

Were they increasing in frequency? Definitely they were. A shower of lines bracketing the scope center substantiated him.

He realized why more tracks appeared near the center than at the edge of the scope. Most particles evidently were small enough so that at the outer limits of the radar's range the trails made no register. Also, the tracks glowed brighter near the center and faded toward the edge.

Too, he became aware the trails were hardly straight. The ship's transit velocity through the Belt bent the trails toward an arc on the scope face.

He saw the track start at the top: but before realization came that it had gone no farther than the center, his head jarred in an instantaneous headache. The quick jolt through his feet and buttocks arrived at the same time, and his sight washed away into a watery blur.

Naturally, after admonishing the crew to use ear plugs, he had neglected to use his. While his eardrums still throbbed with the sharp compression, he fought for clear vision.

The hull mockup illuminated, he searched for the point of impact on the electronic three-dimensional damage guide. No wonder all the rough stuff; it turned out to be a good-sized crater above the control compartment. Perhaps it hadn't been as bad elsewhere. There was no penetration, but after that wallop he wasn't looking forward to any.

"Check in!" he announced.

Dutifully the crew responded, their voices sounding heavy with affected steadiness.

"That landed on the front above control. The party's livening up, so stand by."

Hiller noticed with concern the starfield drifting by the ports. The positional dial showed 17.6 but falling.

"What's with the gyros, Art?" he asked.

"Impact shifted the ship position," he answered. "I'm resetting."

The commander bit his lip, suppressing the pun crossing his mind that this was a new angle. He hadn't figured that much kinetic energy affecting the ship position. As long as the impact came near center, fine; but with a strike near the extremities of the ship, naturally the effect was to spin it, like a top without a molecule of friction.

Oversight Number One. Hell, why count 'em? This one in itself could be fatal. The gyros were never meant to counteract that kind of gyration. Maybe a couple of impacts, yes. After that, they could burn out.

Somebody opened the door of a boiler factory and shut it in a millisecond. The reverberation surprisingly proved slight.

The commander peered closely at the damage guide. A short dark line near the stern: it had taken him a second inspection to find it. He had been looking for a crater.

"Crease over the firing chambers," he reported, then shifted his attention to the indicator. The needle faltered at 18 as the gyros kicked in harder and fell toward 16 again.

"What's the condition of the gyro motors, Art?" Hiller asked.

"Warming up," was the answer, "They're going to have to run full to do any good at all."

"How about using the jets once in a while," Hiller suggested. "Too hard on the fuel?"

"Once in a while, it wouldn't be," the engineer replied. "Constantly steadying a spinning ship this big with the guide jets would take more fuel than we could spare."

Hiller swiftly considered the few possibilities there were. Burning out the gyros was a risk he could not take. Going over the fuel margin was out of the question. And the alternative to these—spinning until they left the Belt.

Spinning provided the only choice. It wasn't necessarily fatal, but it increased the chances for perpendicular strikes. Actually, with such conditions, Bleck's sneering odds held more merit.

Bleck!

The shadow behind him, only a vague outline on the control panel, moved. Hiller fell sideways from the seat, twisting around one of the arms.

Bleck's magnetized boot slammed into the seat and left him overbalancing long enough for Hiller to scramble to his feet.

The man appeared berserk with fear, except he had it channeled toward the destruction of what he assigned as its cause—the ship commander.

No need to search; nothing serving as a weapon lay within reach. Taking the time to stoop and remove his boot meant suicide.

Warily Bleck advanced with the retrieved boot upraised, clumsily limping on the other. Hiller backed until he felt the acceleration straps behind him on the bulkhead. There was no more backing after that.

The last resort—something he did not relish doing—was broadcasting the crew his plight, pulling them from their stations. Anyway, by the time someone arrived—if that didn't faze the man, he would have to try ducking under the weapon and fighting it out.

As Bleck paused to savor his ascendant position and measure the clobbering distance, Hiller started the first word of the announcement. His thinking was riding the crest of a wave of fear which threatened at any moment to break. And the first word was all he managed.

What saved him was his grasp of the straps behind him. On low for movement, his boots would not have held.

His grip had tightened instinctively the moment the ship lurched to the port side, a lurch so sharp he swung out from the bulkhead. His head and chest felt as if they would cave in under the compression.

Wearing only one boot, the other demagnetized, Bleck probably was only beginning to analyze how he was dying when he sailed the length of the control room. His free boot dented the bulkhead and rang against the floor. The boot attached to his foot was hidden under the mixture of sodden clothes and shattered limbs that clung wetly to the bulkhead and began oozing toward the outside of the centrifuge.

For the ship was now gyrating tightly, the stars parading endlessly past the ports. Coming out of shock, strangely, was what bothered Hiller most, the merry-go-rounding.

His hands hurt, he noticed, so he released the needless grip on the straps. Dazedly he navigated to the control seat, sat down, and this time fastened his nylon safety bands and set his boots for high.

The concussion effects wouldn't blink out of his eyes and he stared blearily at the damage indicator. He also found it difficult keeping his eyes from Bleck's remains.

"Fred? Fred!" It was Art's voice. Of course, he hadn't announced damage yet. How long had it been?

"Report!" That's all the commander could get out.

The crew responded weakly. The roll gave him time to locate the damage as a definite penetration in the fuel chambers, evidently by a large particle. The TV monitors showed no tanks dented, and the fine gauges indicated no leaks. One thing, though: the temperature of the tanks had skyrocketed.

He announced the damage and ordered suits on. It felt good to be thinking again. A penetration in the air-filled portion of the ship and the temperature could bake uninsulated flesh promptly. Oversight Number Two.

Art reminded him over the intercom diplomatically, "I'm not counteracting the spin, Fred."

"That's all we can do," the commander returned. "We're going to have to spin through and like it."

"We'll be in the thickest in a couple of minutes." It was Hollender's voice. "I think the patch crew ought to get some business."

"Belt in unless you're traveling," Hiller reminded. Only then did he bring himself to relate to the crew how Bleck died, hoping it would be of constructive value, provided they didn't frighten.

With the next oblique collisions Hiller found the suit better muffled the sound. He wished there was something to be done about the wrenching of his insides at each impact.

The suits helped little on the more direct collisions. Added to that, the ship was gyrating faster and pseudo-gravity pulled at him from the front ports. Giddiness on top of everything else was not improving matters.

He crumpled under the wave of heat and compression when the first particle penetrated the air compartments of the ship. Three of his instrument dials cracked and he felt as if he had received a blow on each square inch of his body. The penetration he located as in the sleeping quarters and sent the patch crew there at once.



About that time the second one penetrated. The jolt was sickening. Somerset reported both members of his crew unconscious when their boots let them slide against bulkheads at the impact. Worse, he said the patch equipment had spun loose and shorted, bent, and fused. He made clear any patch repair as being hopeless.

While Hiller listened to the report, he was sick inside his suit from the centrifugal effect. He recalled how he'd also been sick on the Whirlwind ride at the amusement park when he was a kid. A hell of a space commander. They could use a good collision against the direction of gyration any time, provided the sudden deceleration of the twirl didn't hemorrhage them internally.

Why was he worried about gyrations when the patch kit was a casualty? That latest development cinched it: the odds on getting through were falling every minute. He wasn't facing it, either.

One favorable element, however, was appearing: the particles size remained uniformly small. No structural damage of any consequence had occurred from the collisions already experienced. The hull, at least, could sustain the heat and explosion effects.

Subawarely the commander realized his thinking was punchy. The impacts of missiles against the heating ship's hull constituted a slowly fading pattern of noise and pressure and pain which he was observing objectively, almost amusedly. When he attempted to read the damage indicator or communicate with the crew, the effort became immense and the discomfort great. So much easier to remain contemplative about it.

No doubt this was the condition of the crew. After so much beating, the organic function can tolerate no more. Oversight Number Three.

The commander was aware sufficiently to hope Art Eastburn kept the air cooler circulating. He had already assumed, since the crew was suited in, that the engineer had cut off the fresh air supply. They didn't have to lose it all, just most of it, enough to suffocate somewhere in space.

That hunch? Seemed a hunch fitted in there somewhere. Was it really important? Nothing seemed important except escaping the punishment the particles of the Inner Asteroid Belt were inflicting on the near-senseless bodies in the spinning ship.

His thought processes alternately raced and then froze in a semi-conscious sleep. Between impacts rationality awoke in brief segments of contemplative continuity and slowed when another concussion shuddered the ship. And soon there was no rationality but fantasies rooted in present trauma....

Starlight seeped through the punctured hull around the control chamber. The air supply had long since whistled into space. What ship atmosphere that was salvaged had been piped into the suits and rationed among the men.

They had circumnavigated the Inner Belt after plotting a course back to Earth. Hollender's computations presented them with a rough chance of making it before the air would no longer maintain their life processes.

But it had not worked out. The Earth was yet a bright star in the front ports when the coughing began, when the function of respiration became painful labor.

Some were already choosing the quick way out. Hollender had entered the control room, waved a hand in salute, and unzipped his suit, even as Hiller watched. The instant freezing from the space-filled ship bloated the body slightly, but otherwise there was little difference. Hollender stood statuesquely, coldly rigid, clamped solidly by his boots.

Art Eastburn arrived next, unsmiling. The two men regarded each other, chests heaving, for an endless moment. The mechanical engineer reached for his suit zipper.

"Art, hold on! Not yet, Art, not yet!"

"Not what, Fred? Come out of it, man!"

Eastburn was standing over him, speaking against the plastiglas of Hiller's visor. He sat before the control board, still cinched in his seat. The mechanical engineer wore no suit and he was smiling.

"We're through," his friend was saying. "We made it, Fred."

The ship commander shook his head. The words were supposed to mean something vital. He played them back in his mind.

"We're through. We're through."

If he could understand why the silence hurt his ears, why he was tense, why—Realization spread over his body in a wave of exhilarating relief.

Speech failed him after Art helped him remove his suit. Speech was unnecessary the way Art rapidly filled him in on the lack of casualties and minor damage.

"How long was I out?" the commander at last brought himself to ask, noticing Bleck's body had been removed.

"Over an hour," Art answered. "When the rocks stopped punching I couldn't raise you on the intercom. Found you passed out. You wouldn't revive so I took advantage of my second-in-command rank and straightened out the ship's spin with the guide jets."

Hiller glanced at the ports. The stars rode steadily, and he was aware his viscera felt stable.

"But dammit, Art, all this air!" Hiller complained, waving his hand over his head. "Aren't you over-generous? We must have lost enough through the hull to put us in suits, or at least turn us back."

The engineer grinned teasingly. "I don't think we've lost a cubic inch, Fred."

"The patch kit?"

"Still out."

"But all those penetrations with us in a twirl—"

"All taken care of." Art was enjoying himself.

Hiller's hunch, never considered seriously, jumped back into his mind. That had to be the only explanation.

Art was going on, "As a matter of fact, there's a good example right there." He pointed above them to the bulkhead, layered with plastic, a coolant area, and duralite, that separated the men from space. "One of the toughest hits the ship took, blasted an inch-round hole, looks like. No wonder you conked out."

The after effects of the experience again was making it difficult for the commander to focus his eyes. He unbound his seat bands and clanked directly under the spot, his friend following.

From the closer viewpoint he could see a small, glistening white circle in the bulkhead surrounded by a ring of heat-discolored metal. That was no patch.

He grinned back at Art. "Automatic, eh?"

"I never considered the possibility," Art replied. "I figured the inside pressure would be too great."

"I'm not trying to sound off big," the commander said, "but I had it in the back of my mind when I decided to sail through. As it turned out, it meant the difference between survival or otherwise. Had I known that, I might not have gambled."

Fred Hiller returned to his seat and pushed himself down. His strength was only beginning to return.

"With a bigger hole, it wouldn't have worked. But I was counting on little holes with our strong hull. It would take more pressure than what's inside the ship to stop the instant freeze of space cold in small openings like that.

"I think our frozen air plugs will hold way longer than it takes to repair the patch kit. Matter of fact, I may leave them in until we hit Mars' atmosphere. I'm feeling sentimental about them already!"

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